

Md. Mofrat Shahriar

Executive Engineer (Current Charge)

Bangladesh Rural Electrification Board
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EDUCATION

B.Sc. in Electrical and Electronic Engineering EEE)

Rajshahi University of Engineering & Technology (RUET).

Year of Passing: October, 2016

CGPA: **3.26/4.00** (WES-**3.31/4.00**)

RESEARCH INTERESTS

➤ Power Electronics ➤ Power System ➤ Renewable Energy ➤ Smart Grid ➤ Grid Integration

RESEARCH EXPERIENCES

Undergraduate Thesis on “**Stability Analysis of DFIG and PMSG Based Hybrid Wind Farm Using SMES**”

- Developed a hybrid wind power generation system integrating DFIG and PMSG using MATLAB/SIMULINK.
- Designed a Superconducting Magnetic Energy Storage (SMES) unit for output power stabilization.
- Performed comparative simulations showing significant improvement in power quality and reduction in fluctuations with SMES integration.

PUBLICATION

M. A. A. Khan, M. M. Shahriar and M. S. Islam, “**Power Quality Improvement of DFIG and PMSG Based Hybrid Wind Farm Using SMES**,” *2017 2nd International Conference on Electrical & Electronic Engineering (ICEEE)*, Rajshahi, Bangladesh, 2017, pp. 1-4, doi: 10.1109/CEEE.2017.8412899.

CONFERENCE PRESENTATION

Oral Presentations

- **Md. Mofrat Shahriar** and Tamanna Tasnim Prova.
23rd International Mathematics Conference (**May, 2024**), organized by the Department of Mathematics, University of Rajshahi, Bangladesh.
Title: *Fuzzy Logic Based Energy Management of a Hybrid Power System to Ensure Frequency Stability and Sustainable Green Environment.*

PROJECT

Line Follower and Obstacle Avoiding Robot.

- Supervised by Dr. Md. Selim Hossain, Professor, Department of EEE, RUET.

WORK EXPERIENCES

Executive Engineer (Current Charge)

August 2025- Present

Bangladesh Rural Electrification Board (BREB).

- Lead system protection and design activities, including planning, analysis, equipment coordination, fault investigation, technical standardization, and staff guidance to ensure safe, reliable, and efficient power distribution across Rajshahi Division, Bangladesh.

Assistant Engineer

November 2018 – August 2025.

Bangladesh Rural Electrification Board (BREB).

- Conduct electrical testing of materials (e.g., transformers, cables) in accordance with ANSI and IEC standards.
- Ensure design integrity, maintenance protocols, and safety compliance in rural electrification projects.
- Support quality assurance and technical evaluation for power distribution infrastructure.

Instructor (Electrical)

February 2018 - November 2018.

Borak Polytechnic Institute, Dhaka, Bangladesh.

- Course Taught: Basic Electricity, Basic Electronics, Electrical Circuits, Electrical Circuits Laboratory, Industrial Electronics, Machine, Generation of Electrical Power, Transmission and Distribution of Electrical power.
- Conducted theory and practical classes as per Bangladesh Technical Education Board (BTEB) curriculum for subject-related courses.
- Prepared lesson plans, question papers, worksheets, and academic reports.

WORK EXPERIENCES CONTINUED

Executive (Engineering Department).

November 2017 - February 2018.

SQUARE Herbal & Nutraceuticals Ltd.

- Ensured preventive and breakdown maintenance of process machinery to maximize efficiency and uptime in production units.
- Participated in installation, commissioning, qualification, and validation of equipment, while maintaining engineering documentation and inventory.
- Prepared SOPs/EOPs, conducted staff training, and supervised junior colleagues for consistent operational performance.

TRAININGS

- **Substation Automation System.** 23rd February 2026 – 05th March 2026.
Bangladesh Power Management Institute (BPMI).
- **33/11 KV Indoor Substation Maintenance.** 28th January 2024 – 05th February 2024.
Bangladesh Rural Electrification Board (BREB).
- **Basic Procurement Training.** 22nd January 2022 – 07th February 2022.
Bangladesh Public Procurement Authority (BPPA).
- **Net Metering Guideline-2018 and Solar Irrigation Pump Grid Integration Guideline-2018** 25th February 2021.
Sustainable and Renewable Energy Development Authority (SREDA).
- **Basic Training on Smart Pre-Payment Metering System & Smart Grid.** 25th October 2020 – 08th November 2020.
Bangladesh Power Management Institute (BPMI).
- **Industrial Training Course.** 20th June 2015 – 02nd July 2015.
Ghorashal Thermal Power Station (GTPS- 950MW), BPDB, Bangladesh.

WORKSHOPS

- **Portable Submarine Power Cable Fault Location System.** 27th & 28th January, 2025.
(Basic Operation of BAUR Syscompact 4000)
Iconic Engineering & Trading Company Limited, Dhaka.
- **Basic Operation of TRAX-280.** 14th & 15th June, 2022.
System operation Workshop, BREB, Savar, Dhaka.
- **CT & PT Test and Metering Procedure of BREB.** 11th September, 2021.
System operation Workshop, BREB, Savar, Dhaka.
- **Microcontroller & Robotics.** 6th & 7th June, 2013.
Robotic Society of Rajshahi University of Engineering & Technology (RUET).

TECHNICAL SKILLS

- Programming Language: C Programming, MATLAB.
- Simulation and Design Software: Simulink, Proteus.
- Hardware: Arduino.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- Ambassador of “**IEEE Day 2015 & 2016**” in Bangladesh Section.
- Student Ambassador of “**IEEE madC 2016**” in Bangladesh Section.
- Volunteer of “**Somanupatik**”, a well-known volunteer organization in Bangladesh.
- Technical Adviser of “**Robotic Society of RUET**”, an organization of RUET.
- Participant of “**IEEE Xtreme Programming Competition 9.0**”, held on 24th October, 2015.
- Participant of “**Line Following Robot Competition, Mecceleration**”, held on 19th September, 2014, IUT.
- Participant of “**Line Following Robot Competition, Roborun & Idea Contest**”, held on 27th September, 2013, RUET.

SCHOLARSHIPS

- Education Board Scholarship for Higher Secondary School Certificate Examination. 2011
- Education Board Scholarship for Secondary School Certificate Examination. 2009
- Junior Scholarship 2006
- Primary Scholarship 2003

MEMBERSHIP

IEEE Young Professionals and Graduate Student Member

Membership No- 9289773, R10 -Asia and Pacific.